



Divyam Jain

Roll No. 220107035

B.Tech. Chemical Engineering

Indian Institute of Technology Guwahati

+91-6280109478

j.divyam@iitg.ac.in

divyamjainiitg@gmail.com

Portfolio | divyamiitg | divyamiitg

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology Guwahati	8.24	2022-Present
Senior Secondary	Central Board of Secondary Education	93%	2022
Secondary	Central Board of Secondary Education	95%	2020

EXPERIENCE

- Summer Associate** | Nation with NaMo Jun 2025 - Jul 2025
*Received **Pre-Placement Offer** owing to exceptional internship performance* LoR
 - Presented a **5-year** roadmap to PMO for **360° welfare** program benefiting **3M+** **ex-servicemen** of Indian Armed Forces
 - Led financial security module making **20+ entitlements** equitable across services & **10 ranks** via consolidated framework
 - Analysed insights from 50+ user interviews and **25+ veteran programs** across 5 countries, predicting **15+ policy gaps**
 - Strategised **10 policy** recommendations amounting to INR 2500 cr +, ensuring fiscal prudence with **2% budget** increase

PUBLICATION

- Chemical Absorption of Carbon Dioxide in Taylor Flow along a Curved Microchannel**
Guide: Prof. Raghvendra Gupta | Accepted at 2nd edition of ICOM
 - Built an **OpenCV pipeline**, implementing **stateful tracking model (ROI)** with sub-pixel refinement for 30,000+ frames
 - Applied **Polynomial Regression** to time-series data **correlating process inputs** with bubble dynamics across 4 channels

PROJECTS

- Natural Language to SQL Chatbot using LangChain** Jan 2025
Techevince 11.0 (IIT Guwahati's Tech Expo) - Judges' Choice (Business category) GitHub | Demo
 - Deployed **AI agent** on **MySQL** database (**8 tables, 100k + records**), automating analytics and saving **90% query time**
 - Improved **accuracy** by **40%** with semantic query search and adaptive table selection using **LangChain** and **ChromaDB**
 - Enabled real-time decision making via **Streamlit** prototype powered by **Gemini API** connected to **Aiven** cloud database
- Credit Card Default Prediction for IDFC First Bank** Feb 2024
Pan-IIT ML Hackathon (Convolve 2.0) - Top 10 / 1500 teams Report
 - Developed credit default risk model (**140k+ records**) using ensemble of 15 **XGBoost**, **CatBoost** and **LightGBM** models
 - Captured **90% variance** in credit default behavior, engineering **282 features** via **AutoFE**, **PCA**, **RFE** with **100-fold CV**
 - Reduced **high-risk false negatives** optimising recall using **NearMiss** undersampling, **logit shift**, and **Youden's J** thresholds
- Pairs Trading and Cointegration Strategy** Dec 2024
Finance and Economics Club, IIT Guwahati GitHub
 - Developed a **low-frequency** trading strategy using **Co-integration principles**, identifying statistically similar stock pairs
 - Utilized **Z-Scores** to generate trading signals rooted in mean reversion, predicting **profitable relative price** movements
 - Leveraged **Augmented Dickey-Fuller test** to validate **stationarity** of the spread & analysed Sharpe ratio and Drawdown
- Abstractive Text Summarisation using RNN and Transformers** June 2024
Consulting & Analytics Club, IIT Guwahati GitHub
 - Built **GRU seq2seq** models with **attention** mechanisms and subword tokenization to generate abstractive text summaries
 - Fine-tuned T5 & BART encoder-decoder models with **beam-search decoding** producing coherent and concise outputs
 - Evaluated models using **ROUGE** & **BLEU** metrics, analyzing trade-offs between coverage, conciseness, linguistic quality

TECHNICAL SKILLS & KEY COURSES TAKEN

- Programming:** C, C++, Python
- Data Science:** NumPy, Pandas, Scikit-learn, TensorFlow
- Mathematics:** MA 102 (Probability, Statistics, Real Analysis)
- Web Technologies:** HTML, CSS, Streamlit
- Data Visualization:** Tableau, Matplotlib, Seaborn
- MOOC:** Summer Analytics, Time Series Analysis

POSITIONS OF RESPONSIBILITY

- Head of Consulting**, Consulting & Analytics Club, IIT Guwahati Apr 2024 - Mar 2025
*Awarded the **Best Technical Club** among 15+ clubs under Students' Gymkhana Council*
 - Spearheaded a team of **100+** associates & **25** core members organising **10+** international events (like Krack the Case)
 - Directed a 5-week bootcamp on Business Strategy, mentoring **15000+** learners (**70% YoY** growth) from **30+** countries
 - Authored IIT Guwahati's **first-ever** Business casebook and Product guide impacting **100k +** learners within **3** months
 - Secured **INR 3 L** (**150 % YoY** growth) from **15+** sponsors organising Strategy Storm, an international case competition

ACHIEVEMENTS & EXTRACURRICULARS

- Bronze Medal**, Energy Consulting challenge at **Inter IIT Technical Meet 13.0** - IIT Bombay among 21 IITs 2025
- Silver Medal**, Product Development challenge at **Inter IIT Technical Meet 12.0** - IIT Madras among 21 IITs 2024
- Bronze Medal**, Cascade Cup 2nd edition **Machine Learning Hackathon** among **2000+** participants 2024
- Gold Medal**, Inventory Optimisation challenge at **Inter-Hostel Tech Competition** - Kriti among **14** hostels 2024
- Global Top 5**, Equity Research Challenge among **2000+** participating teams from 500+ institutes. 2024
- Coordinated as **Corporate Relations** executive in north-east India's largest techno-management fest **Techniche** 2024
- Top 100 rank**, in National Talent Search Examination (NTSE) - Stage 1 among **50,000+** candidates in Punjab 2020
- Top 300 rank**, in Indian Mathematics Olympiad (IMO) - Qualifiers among **10,000+** candidates in North region 2020